

ES-MEGA-ALU-ENERGY-SYSTEM

Full Technical Documentation (English)

1. Executive Summary

The ES-Mega-Alu Energy System is a modular, emission-free large-scale power plant that generates energy through the controlled electrochemical reaction of aluminum and water. It combines direct electricity generation, hydrogen production, and process heat within a fully closed material cycle.

Three architectural layers:

1. Standard Cistern (Module)

Vertical reaction chamber with aluminum plate stacks, precise electrolyte flow, active thermal management, and defined H₂ extraction.

Output: **20–24 MW per module.**

2. Cluster (50 modules)

Autonomous energy block with shared circuits for electrolyte, hydroxide, hydrogen, and heat.

Output: **600 MW direct electric, + 350 MW hydrogen-electric, + 200 MW thermal.**

3. Mega Power Plant (500+ modules)

Industrial-scale energy ecosystem with **6–11 GW total output**, including cistern fields, recycling facilities, hydrogen center, heat network, and control unit.

The system is emission-free, silent, safe, scalable, and fully circular.

2. Reaction Fundamentals

Reaction:

Aluminum + Water + Electrolyte → Electrons + H₂ + Heat + Aluminum Hydroxide

Energy paths:

- Direct electricity: 3.8–4.2 kWh/kg Al
- Hydrogen electricity: 1.8–2.2 kWh/kg Al
- Thermal: 1–1.5 kWh/kg Al

Total efficiency: 70–85 %

3. Standard Cistern (Root Module)

3.1 Structure

- Vertical reaction chamber
- Volume: 40–80 m³
- Ceramic/polymer lining
- Reinforced concrete shell
- Top access, side inspection, bottom drain

3.2 Power Stack

- Plates: 1.0 × 1.5 m, 10–30 mm
- Micro-textured surface
- Spacing: 8–12 mm
- 100–150 plates
- 150–225 m² active area

3.3 Electrolyte Flow

- Bottom inlet with diffuser
- Laminar upward flow
- Flow rate: 8–15 m³/h
- Continuous hydroxide removal
- Gas separator at outlet

3.4 Thermal Management

- Optimal: 40–70 °C
- Heat exchangers: 100–300 kW
- Sensors: bottom, mid, top, gas space, return line

3.5 Hydrogen Handling

- Gas headspace
- H₂ separator
- Check valves
- Storage or fuel cells

3.6 Output

- 12 MW direct electric
 - 5–7 MW hydrogen-electric
 - 3–5 MW thermal
 - Total: **20–24 MW**
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4. Cluster (50 Modules)

- Shared circuits
 - Central control
 - Output: **1.1–1.2 GW**
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5. Mega Power Plant (500+ Modules)

- 6 GW direct electric
 - 3 GW hydrogen-electric
 - 2 GW thermal
 - Total: **11 GW**
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6. Operational Cycle

1. Pre-start
 2. Initial flooding
 3. Ramp-up (0–100 % in 60 s)
 4. Steady operation
 5. Three energy paths
 6. Material cycle
 7. Load control
 8. End of cycle
 9. Shutdown
 10. Reset
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End of English Documentation